A growing epithelial spheroid in an extracellular matrix: the Plexus2 implementation

`prototype/ecm` · runs in `log/okuda_ECM` · 6 August 2026

1. Sets, operators, schedule

A Plexus model is sets × fields × operators × schedule. The biological hierarchy here is four entities and one field; the basement membrane is a *specialised* ECM, not the ECM:

entity	set	representation	principal operators
epithelium	`vertex`, `cell`	3D active vertex model	`shape_energy_3d`, `morphogen_growth_3d`, `di`
junction	relation on the mesh	half-edges, keyed state	`junction_myosin`, `reconnect_t1_3d`
basement membrane	`basement_membrane_particle`	MPM + explicit bonds	`..._seed/bond/remodel/bond_break`, `integr`
interstitial ECM	`mpm_particle`	MPM fibres	`ecm_seed`, `ecm_stress`, `cell_to_ecm`
(field)	`mpm_grid`	MLS-MPM background grid	`mpm_scatter/gather/grid_update`

Every particle body scatters into and gathers from the *same* grid; that sharing *is* the mechanical coupling between them, so no pairwise contact model is written. Operators carry one of seven kinds. The four dynamics kinds return a per-set delta the engine integrates once per tick; `field`, `rewire` and `structural` mutate the grid, the relation E, or the membership $|S|$ in place and *return nothing.* That last fact drives a design decision in §5.

2. The spheroid: 3D active vertex model

Cells are faces of a closed triangulated shell; cell f's volume is the wedge from the tissue centre, $v_f = \frac{1}{3}\,\mathbf{c}_f\cdot\mathbf{N}_f$. The energy is

$$E \;=\; \sum_f \Big[K_A\,(A_f - A_f^0)^2 + K_P\,(P_f - P_f^0)^2 + \tfrac{1}{2}\Gamma P_f^2 + K_V\,(v_f - v_f^0)^2\Big] \;+\; \Lambda\sum_e m_e\,\ell_e \;+\; K_R\sum_i \big(|\mathbf{x}_i| - R_0\big)^2, \quad (1)$$

relaxed overdamped each frame, $\mathbf{x} \leftarrow \mathbf{x} - \eta\mu\,\nabla E$ with a per-step cap. The factor m_e is new (§5); with $m_e \equiv 1$ Eq. (1) is the stock Okuda energy and every prior run is bit-identical. Growth scales each cell's targets by a cumulative s_f,

$$s_f \leftarrow s_f\big(1 + \lambda(\rho + \mathrm{Hill}(a_f))\big), \qquad A_f^0 = A_f^{0,\mathrm{init}} s_f^2, \quad P_f^0 = P_f^{0,\mathrm{init}} s_f, \quad v_f^0 = v_f^{0,\mathrm{init}} s_f^3, \quad (2)$$

and `divide_3d` splits a cell once v_f passes twice its reference. From 200 cells the tissue reaches ~ 6000 and its apical radius $4.66 \to 16.6$ (arbitrary units) in 402 frames.

3. The ECM: MLS-MPM

The matrix is material points on a background grid, fixed-corotated with $F = RS$:

$$\boldsymbol{\tau} \;=\; 2\mu\,(F - R)F^{\top} + \lambda J(J-1)\,I, \qquad \boldsymbol{\sigma} \;=\; \boldsymbol{\tau}/J, \qquad \sigma_{\mathrm{vM}} = \sqrt{\tfrac{3}{2}\,\mathbf{s}\!:\!\mathbf{s}}, \;\; \mathbf{s} = \boldsymbol{\sigma} - \tfrac{1}{3}\mathrm{tr}(\boldsymbol{\sigma})I. \quad (3)$$

$\boldsymbol{\tau}$ was already computed every substep to build the affine momentum matrix and then discarded; `mpm_scatter` now optionally caches the Cauchy stress $\boldsymbol{\sigma}$ to a per-particle buffer (`store_stress`, off by default, byte-identical when off). This matters quantitatively: colouring by the volumetric measure $|J-1|$ put 0.6% of the matrix above the display floor, whereas σ_{vM} of the *same* run puts 64% there. A matrix pushed by a growing sphere *shears*; the fixed-corotated law resists volume change far more stiffly than shape change, so the volumetric measure was blind to the deformation, not the mechanics.

4. Spheroid ↔ ECM

The two solvers cannot share a world, and the reason is not a software limit. `mpm_grid` is hard-coded to $[0,1]^3$ with $dx = 1/n_{\mathrm{grid}}$, while the epithelium lives in a 50-unit box seeding cells at radius 5. Rescaling the tissue into the unit box was tried and measured, not assumed: the same run gave $200 \to 319$ cells at its own scale, $200 \to 212$ under a pure geometric rescale (it *collapses*), and $200 \to 201$ under a dimensional one—stable, but no longer growing. The cause is that the terms of Eq. (1) carry different powers of length, $K_A(\Delta A)^2 \sim s^4$, $K_P(\Delta P)^2 \sim s^2$, $K_V(\Delta V)^2 \sim s^6$, $\Lambda P \sim s$, so a 50× shrink changes *which term wins* and surface tension takes over; correcting every exponent stops the collapse and still does not restore growth. Calibrating a vertex model to a new scale is a project, and doing it badly would mean running the ECM experiment against a tissue that is no longer the reference one while the movie looked right. So the two are run as two passes joined by a single geometric scale s, applied to a *recorded* shape rather than to a simulation. The tissue surface is recorded as an angular radius map $R(\theta,\phi,t)$ over its apical vertices. Contact is a one-sided penalty plus a hard projection,

$$\mathbf{a}_i = k\,\big(R(\hat{\mathbf{u}}_i) - r_i\big)\,\hat{\mathbf{u}}_i \;\text{ for }\; r_i < R, \qquad \text{then} \qquad \mathbf{x}_i \leftarrow \mathbf{c} + \hat{\mathbf{u}}_i\,R(1+\varepsilon) \;\text{ if still inside,} \quad (4)$$

the projection (`cell_exclude_3d`) being necessary because a penalty punishes penetration rather than preventing it. The reaction is accumulated by direction into a pressure map and normalised by the solid angle each bin subtends, $P(\theta, \phi, t) = \sum_{i \in \text{bin}} k\, \text{depth}_i \,/\, (R^2\, d\Omega)$, which closes the loop back onto the cells: `ecm_growth_gate_3d` slows the cell cycle where the matrix presses,

$$g \;=\; f_0 + \frac{1 - f_0}{1 + (P/P_{1/2})^n}, \qquad v_f^0 \leftarrow v_f^{0,\text{prev}} + g\,\big(v_f^{0,\text{now}} - v_f^{0,\text{prev}}\big), \tag{5}$$

i.e. the growth *increment* of Eq. (2) is gated, so a loaded cell reaches its division threshold later and divides less often. This is a rate effect and compounds: it turns a 2.75× stress anisotropy into a shape. Measured, against a round control at oblateness 1.009: gating on a polar stress pattern gives 1.32–1.43, while a *flat* pattern of the same strength shrinks the tissue just as hard and leaves it round (1.015). The shape comes from the pattern, not the amount.

That last sentence is now known to be an identity rather than a finding, and the correction matters more than the claim did. The gate reads the map normalised by its own p99 against a $p_{1/2}$ that is the late-run median in the same units, so it depends only on $P/\text{median}(P)$: **multiplying the whole pressure field by any constant leaves it unchanged.** The invariance was built deliberately—so $p_{1/2}$ would mean the same thing whatever the matrix stiffness—and the null of the stiffness sweep was then reported as a discovery. Two further constraints are also imposed rather than emergent: `smooth_phi_deg`= 360 makes the map axisymmetric before any cell divides, so the outcome can only be a spheroid of revolution; and the estimator's own null on a perfect sphere is 1.020, so every oblateness below ≈1.05 quoted anywhere in this work sits at or under it. What the campaign actually shows is that *a prescribed polar growth mask makes an oblate tissue.* Whether matrix stress is what sets the mask is untested, and the Hill exponent $n = 6$ lies outside the $n \in [1, 2]$ that the one published calibration of this functional form allows.

5. The two new levels

Junction myosin. Actomyosin previously existed only as the scalar Λ of Eq. (1). It is now per-junction, recruited by tension and relaxing over τ_m:

$$m_e^{\text{ss}} \;=\; \alpha\Big(1 + \beta\big(\ell_e/\langle\ell\rangle - 1\big)\Big), \qquad \frac{dm_e}{dt} \;=\; \frac{m_e^{\text{ss}} - m_e}{\tau_m}, \tag{6}$$

with α the global activity—the *in silico* myosin inhibition. Because `rewire` and `structural` operators return no delta (§1), any state living on junctions is silently invalidated by every T1, division and death. So m_e is keyed by *topological identity*, the unordered vertex pair $(\min(v_i, v_j), \max(v_i, v_j))$: a surviving edge finds its own value, a new edge takes m_{new}, a deleted edge stops being asked, and `reconnect_t1_3d` and `divide_3d` need no edits. A test battery confirms α=1, β=0 is bit-identical to no operator, that the radius falls monotonically over $\alpha \in [0, 3]$ (7.22 → 6.14), and that new junctions appear at 10.5 per division rather than at the edge count per frame—the signature that the keying holds.

Basement membrane. A monolayer of material points seeded on $R(\hat{\mathbf{u}}, 0) + \delta$ and joined by explicit *crosslinks*, so that the sheet can be *defective*: MPM particles are independent material points coupled only through the grid, so an MPM-only membrane has no connectivity, cannot be crosslinked and cannot fragment. A crosslink is a two-body bond (i, j) between neighbouring membrane particles, built once by a radius search on the seeded shell and carrying its own rest length ℓ_{ij}^0—it stands for the collagen IV network that carries the load (laminin and nidogen contribute little to stiffness). Adhesion is a different object entirely: a *one-body* spring between a membrane particle and a moving point on the epithelial surface. Four terms:

$$\text{crosslink:}\quad \mathbf{f}_{ij} = k_b\,(\ell_{ij} - \ell_{ij}^0)\,\hat{\mathbf{d}}_{ij}, \qquad \text{turnover:}\quad \ell_{ij}^0 \leftarrow \ell_{ij}^0 + \frac{\ell_{ij} - \ell_{ij}^0}{\tau_r}, \qquad \text{failure:}\quad \frac{\ell_{ij}}{\ell_{ij}^0} - 1 > \varepsilon_b \Rightarrow \text{bond removed}, \tag{7}$$

$$\text{integrin:}\quad \ddot{\mathbf{x}}_i = k_a\big(\mathbf{a}_i(t) - \mathbf{x}_i\big) - c_d\,\dot{\mathbf{x}}_i, \qquad \mathbf{a}_i(t) = \mathbf{c} + \hat{\mathbf{u}}_i\big(R(\hat{\mathbf{u}}_i, t) + \delta\big), \qquad c_d = 2\sqrt{k_a}\ (\zeta = 1). \tag{8}$$

Do the two matrix layers interact? Only by pushing, and only as far as the grid resolves. Both particle bodies scatter into the one `mpm_grid` and gather back from it (`mpm_scatter[accumulate]`), so they exchange momentum—but no bond joins a membrane particle to a fibre, and the two layers slide past one another freely. The missing object is collagen VII: anchoring fibrils staple the basement membrane to the interstitial matrix, and their operator (`basement_to_ecm`) is designed and not written. The quantitative caveat is sharper. At $n_{\text{grid}} = 48$ the cell size is $dx = 0.021$, against a membrane thickness of 0.002 and an in-plane particle spacing of ≈ 0.005: one grid cell holds some sixteen membrane particles and is ten times thicker than the sheet. The grid does not resolve the membrane, it sees a smeared density, and any fibre within one cell of it is effectively co-moving. The strength of the membrane–matrix coupling is therefore set by dx rather than by any parameter of the model, and no membrane–matrix result should be read off this configuration without a resolution study.

Bond removal is registered `rewire`: fragmentation *is* a change of E. What is reported is not the bond count but the largest connected component—a sheet that has lost a third of its crosslinks and is still one piece is not fragmented.

The anchor $\mathbf{a}_i$ of Eq. (8) deserves stating precisely, because it is what distinguishes adhesion from contact. $\hat{\mathbf{u}}_i$ is the particle's *own* direction, frozen at seeding, so the anchor sits at a fixed *angular* position and the spring resists tangential sliding—anchoring instead to the nearest surface point would let the particle slide freely, which is contact, and the shared grid already provides that. The radial coordinate $R(\hat{\mathbf{u}}_i, t)$ follows the epithelium outward, so the bond is radially compliant and tangentially stiff, as an integrin–focal adhesion complex is. Two consequences follow. Geometrically, a particle pinned to a fixed angular position on a surface whose radius triples must accommodate area growing as R^2, so its crosslinks stretch by $\sim R$: this is the loading a basement membrane experiences under tissue growth, and without it the sheet merely slides and its bonds never strain at all (measured: mean strain 0.0000 at every bond stiffness). Mechanically, because the tissue is a replay, $\mathbf{a}_i$ is a *prescribed kinematic target* and not a mutual force pair—the epithelium feels no reaction. That is the one-way coupling of §4 again, and it bites hardest here, since integrin adhesion is precisely the route by which a stiff membrane would restrain a growing acinus.

Two calibration facts, both found by tests rather than by inspection. (i) Writing the crosslink force as $k\,\varepsilon = k(\ell - \ell^0)/\ell^0$ makes it $1/\ell^0 \approx 450\times$ stiffer than the number supplied, and the sheet tore itself apart in 40 frames; Eq. (7) is Hookean in the *extension*, with the explicit-integration bound $\Delta t_{\mathrm{sub}} < 2/\sqrt{k_b}$. (ii) The anchor in Eq. (8) *moves*, at $v_a \approx 0.135$ box/time. An undamped spring does not track a ramp, it oscillates about it with amplitude $v_a/\omega = 9.5 \times 10^{-4}$, against a bond rest length of 2.2×10^{-3}: 43% strain oscillation on every particle, and 66 680 of 70 129 crosslinks broken. At $\zeta = 1$ the steady lag $2v_a/\omega$ is larger but *coherent* between neighbours, so it does not stretch bonds—0 broken, the sheet intact at 10% strain.

A result falls out of Eq. (7) without τ_r: sweeping ε_b over 0.05/0.20/0.60 destroyed the sheet at every value, because the surface grows 66% in linear strain by frame 150 and 260% by the end. A purely elastic basement membrane *cannot* enclose a growing epithelium; remodelling is not optional, and fragmentation becomes a race between τ_r and growth.

Point (ii) above, and everything else in this section about oscillation, critical damping and tracking lag, describes a sheet that was given a **mass**. At $\mathrm{Re} \sim 10^{-10}$ the equation of motion is $\gamma\dot{\mathbf{x}} = \mathbf{F}$, not $m\ddot{\mathbf{x}} = \mathbf{F}$: overdamped there is no oscillation to damp, no lag to critically damp away, and no stability ceiling of the kind measured at $k \approx 8 \times 10^3$. Those were reported as calibration facts found by testing; they are numerical facts about a term that should not have been in the model. The membrane is now integrated overdamped, and the runs that re-measure the ceiling, the sinking and the tearing without inertia are in progress.

Two further corrections to this section. The fragmentation metric it appeals to—the largest connected component—was `NaN` on every frame any analysis read, so it had never once been reported; it is now carried forward, alongside the mean degree z, for which central-force rigidity percolation in 2D requires $z \approx 4$. And `integrin_adhesion` is named for a bond it does not represent: in mammals $\alpha_1\beta_1$ and $\alpha_2\beta_1$ do bind collagen IV, so the receptor exists—but the dominant epithelial anchor is to *laminin*, through $\alpha_3\beta_1$ and $\alpha_6\beta_4$, and laminin is not modelled at all.

6. The two schematics

The two levels added on top of the spheroid and the matrix. Everything else about them — the configurations, the measurements, and the external audit that found much of it wanting — has moved to `spheroid_ecm_archive.tex` while the model is being corrected.

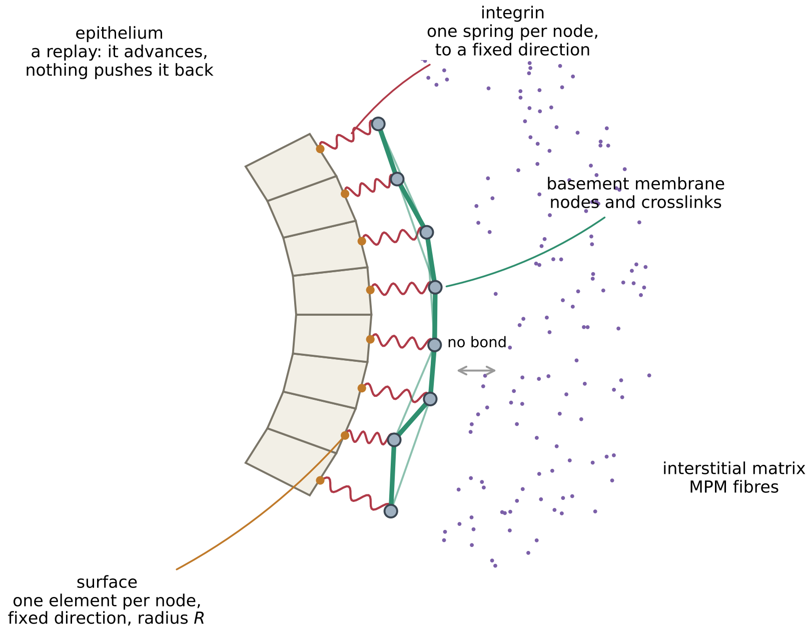

Figure 1: **What the membrane is made of.** Four objects, and the arrows between them are the whole model. The *epithelium* (cream wedges) is a recording: it advances on a prescribed schedule and nothing can push it back. Its outer boundary is the `surface` Level (orange), one element per membrane node, each holding a fixed direction $\hat{\mathbf{u}}_i$ and the radius $R(\hat{\mathbf{u}}_i, t)$ the tissue reached in that direction. The *basement membrane* is grey *nodes* joined by green *crosslinks*, sitting a distance δ outside. Each node is tied to its *own* surface element by one red *integrin* spring—one per node, to a fixed angular position, which is what makes it adhesion rather than contact. Outside is the interstitial matrix (purple), which the membrane never bonds to: they meet only through the shared grid, or in the spring-graph version, not at all.

Figure 2: **What a junction is, and the one loop that makes it interesting.** Left: two cells share an edge, and that edge *is* the junction—it belongs to neither cell, which is why its state is keyed by the unordered pair of vertices (i, j) rather than stored on a cell. Myosin m_e sits on the shared edge and sets its tension. Right: the feedback. A junction longer than average recruits more myosin (β), which pulls harder, which shortens it. That loop is the only thing here that Okuda's model cannot already do.